ECS 240 Project Proposal

Rajiv Choudhury, Jacob Hamzawi, Rishika Nigam, Gurpreet Singh

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1 Introduction and Motivation

1.1 Problem Domain and Motivation

A *side effect* is a modification of state that outlives a function call, that is, state the function did not create itself. Side effects make code difficult to reason about: at a call site it is unclear which components a function may modify, so a reader must inspect the callee and everything it calls[1]. Functional programming avoids this problem and allows for greater modularity[2], and Backus argued for liberating programming from assignment and mutation altogether[3]. Yet effects are often necessary, as with file I/O. Haskell confines them to the monad[4], and effect systems track them statically in types[5,6]. Our domain is *making side effects explicit in an imperative systems language*: every function is pure unless declared `impure`, an impure function must list the outside state it may modify, and the compiler checks the declaration.

1.2 Proposed Language

We propose a toy programming language similar to traditional systems programming languages like C and Rust with the distinction that every function is explicitly defined as "pure" or "impure", where a "pure" function may only modify state initialized in or moved into its scope. Our language will implement at least the minimum set of elements that allow us to illustrate this system, including variables, functions, references, and an 8-bit unsigned integer primitive as the sole type. We may support other features such as structs, strings, or simple file I/O depending on our timeline (see §§ 2 and 4).

Rust itself originally sought such a design, defining all functions as "pure" by default and employing the `io` keyword to explicitly designate a function with I/O side effects.[7] The language designers eventually opted to remove `io` keyword and the pure/impure separation. We hope to discover while designing and analyzing our own language what explicit declaration of side effects could enable for language users and whether the Rust developers' decision was ultimately justified.

```
fn pure_function(x, y) {
    x + y
}

// global variable
var z;

// You must specify the data out of scope whose
// state may change
fn impure(z) impure_function(x, y) {
    z = x + y;
}

// If variables have fields, those fields
// must be declared
fn impure(x.value, y.value) impure_function_struct(x, y) {
    // mutates x
    x.value = 1;
    y.value = 1;
}

// You can look at the program see a full list
// of all the things that could change state
fn impure main() {
    var x = 1;
    var y = 2;
    x = pure_function(x, y);
          // mutated: z
          impure_function(x, y);
}
```

1.3 Proposed Tool

compiler, where the input is a string in our source language and the output is a string in Rust. The user interacts with it from the command line As a stretch goal (see §2), an analysis tool to glean information from impure declarations may be in scope.

1.4 Scope and Related Work

Haskell has the Monad[4] Functional programming allows for greater modularity.[2] And they also often discourage side effects by the nature of their design. `cargo-scan` is a static analysis tool for Rust that detects side effects from Rust source code.[8]

2 Goals

Minimum goal. Parser, interpreter, and working compiler. Interpreter and compiler must detect whether a function is pure or impure and error if a function is impure without declaring so along with the outside state it modifies.

Target goal. The above, along with a static analysis tool that enumerates all the possible state that is modified given a function (including the functions it itself calls). The tool will at least be a CLI, but could be extended to a VSCode extension as well.

Reach goal. The above, as well as other Rust features that might make the division harder and mitigation: system state (file I/O), structs, etc.? Specifically for file I/O, we could explore options to avoid Rust's issue of `io` keyword explosion (similar to Haskell's Monad).

3 Syntax and Semantics

Syntax. The syntax of our language can be described by the following example and the subsequent draft formal grammar:

```
fn pure_function(x, y) {
    x + y
}

// global variable
var z;

// You must specify the data out of scope whose
// state may change
fn impure(z) impure_function(x, y) {
    z = x + y;
}

// If variables have fields, those fields
// must be declared
fn impure(x.value, y.value) impure_function_struct(x, y) {
    // mutates x
    x.value = 1;
    y.value = 1;
}

// Data reached through a reference is out of scope,
// so writing through it must be declared
fn impure(*r) increment(&r) {
    *r = *r + 1;
}

// You can look at the program see a full list
// of all the things that could change state
fn impure main() {
    var x = 1;
    var y = 2;
    x = pure_function(x, y);
    // mutated: z
    impure_function(x, y);
    // mutated: x (through the reference)
    increment(&x);
}
```

Start symbol P (program). Metavariables: n ranges over integer literals 0–255, x, f, ℓ range over identifiers (letters, digits, and underscores, beginning

with a letter) applying to variable, function, and field names respectively.

$$
\begin{aligned}
P &::= D \mid D\ P \\
D &::= \texttt{var}\ x\texttt{;} \mid \texttt{fn}\ a\ f\texttt{(}q_1, \ldots, q_k\texttt{)}\ \texttt{\{}\ B\ \texttt{\}} \\
q &::= x \mid \texttt{\&}x \\
a &::= \epsilon \mid \texttt{impure} \mid \texttt{impure(}p_1, \ldots, p_m\texttt{)} \\
B &::= s_1 \ldots s_j \mid s_1 \ldots s_j\ e \\
s &::= \texttt{var}\ x\texttt{;} \mid \texttt{var}\ x\ \texttt{=}\ e\texttt{;} \mid p\ \texttt{=}\ e\texttt{;} \mid f\texttt{(}e_1, \ldots, e_k\texttt{);} \\
p &::= x \mid p\texttt{.}\ell \mid \texttt{*}p \\
e &::= n \mid p \mid \texttt{\&}p \mid e_1 \oplus e_2 \mid \texttt{(}e\texttt{)} \mid f\texttt{(}e_1, \ldots, e_k\texttt{)} \\
\oplus &::= \texttt{+} \mid \texttt{-} \mid \texttt{*} \mid \texttt{/}
\end{aligned}
$$

where $k, m, j \geq 0$. A place p names a location that can be read, assigned, or declared in an `impure` clause: a variable, a field of a place, or the target of a reference. `&`p creates a reference to p; all references are mutable (Rust's `&mut`) but are written `&`. Function signatures show which arguments are passed by reference with an explicit symbol `&`x. The grammar for e is ambiguous as written: unary `*` and `&` bind tightest, then `*` and `/`, then `+` and `-`, and all binary operators are left-associative. A trailing expression e with no semicolon is the function's return value, as in Rust.

Semantics. The semantics of our language can be described as the set of `(INPUT, prog, OUTPUT)` triples produced by our reference interpreter, which serves as the oracle for the compiler: for every program, the emitted Rust must produce the same output. Programs take no input, so `INPUT` is always empty. This could change if we decide to investigate file I/O or read from stdin. As in Rust, a program that fails the purity check is rejected before it runs and so belongs to no triple. For every other program, `OUTPUT` is the final values of all global variables once `main` returns, or an error if execution faults (e.g., `u8` overflow or division by zero). Our language's purity check must guarantee that in any accepted program, a call to a function declared `impure(`$p_1, \ldots, p_m$`)` changes only $p_1, \ldots, p_m$, and a call to a pure function cannot change state outside its own scope. The interpreter can verify this by comparing state before and after each call, so any violation exposes a bug in the purity check itself. Should our purity check extend to system state (see

§2), its guarantees would extend along with it (e.g. no file is read or written unless explicitly declared, etc.).

4 Implementation Plan

4.1 Tool and Language Selection

Parser in Rust; meta language is Rust; target language for compiler is Rust.

4.2 Incorporating Existing Tools

From scratch.

4.3 Architecture

Core language definition; parser that outputs core; interpreter reads core and executes; compiler reads core and outputs IR, in IR we do checks on purity, then IR to target Rust

4.4 Development Effort

4.5 Challenges

Tracking scope is hard; tool is too much work... Fallback..? What if a variable will *never* change state? if we have if statements...should we detect that? need it be declared?

5 Collaboration Plan

Division of Work. Work divided as follows

Component	**Group Member(s)**
Core	Jacob Hamzawi
Parser	?
Interpreter	?
Compiler	?
Analysis tool	?

Communication. Slack and GitHub; Meetings after class on Monday and over Zoom on Thursday

Timeline. The below timeline assumes a 4-week time allotment to complete the project. It may be contracted or expanded depending on our eventual project due date.

Week	Components
Week 1	Parser
Week 2	Interpreter + Compiler
Week 3	Analysis tool + exploration
Week 4	Presentation

Commitment statement.

- I commit to work and contribute equally to this project. — *Jacob Hamzawi*

6 Conclusion and Questions for Feedback

Re-state goal kinda: does explicit knowledge of impure functions change programming paradigm? Or something like that.

Questions: are variables enough? Should we focus instead on system state? That's more difficult but perhaps more compelling.

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A Complete Formal Grammar (In Progress)

$$
\begin{aligned}
S &::= \\
\texttt{<Int>} &::= 0 \mid \texttt{<NonZeroInt>} \\
\texttt{<NonZeroInt>} &::= \texttt{<NonZeroDigit>} \mid \texttt{<NonZeroInt>} \mathbin{+\!\!+} \texttt{<Digit>} \\
\texttt{<NonZeroDigit>} &::= 1 \mid 2 \mid 3 \mid 4 \mid 5 \mid 6 \mid 7 \mid 8 \mid 9 \\
\texttt{<Digit>} &::= 0 \mid \texttt{<NonZeroDigit>} \\
\texttt{<Exp>} &::= \texttt{<MultDivExp>} \mid \texttt{<Exp> <AddSubOp> <MultDivExp>} \\
\texttt{<AddSubOp>} &::= + \mid - \\
\texttt{<MultDivExp>} &::= \texttt{<Factor>} \mid \texttt{<MultDivExp> <MultDivOp> <Factor>} \\
\texttt{<Factor>} &::= \texttt{<Int>} \mid \texttt{<Var>} \mid \texttt{(<Exp>)} \\
\texttt{<MultDivOp>} &::= * \mid / \\
\texttt{<Body>} &::= \texttt{<StmtList>} \mid \texttt{<Return>} \mid \texttt{<StmtList>} \mathbin{+\!\!+} \texttt{<Return>} \\
\texttt{<StmtList>} &::= \texttt{<Statement>} \mid \texttt{<StmtList>} \mathbin{+\!\!+} \texttt{<Statement>} \\
\texttt{<Statement>} &::= \texttt{<Declaration>;} \mid \texttt{<Assignment>;} \\
\texttt{<Declaration>} &::= \texttt{var <Var>} \mid \texttt{var <Var> = <Exp>} \\
\texttt{<Assignment>} &::= \texttt{<Var> = <Exp>} \\
\texttt{<Return>} &::= \texttt{<Exp>} \\
\texttt{<Call>} &::= \texttt{<Var> = <Exp>} \\
\texttt{<Fn>} &::= \\
\texttt{<FnSig>} &::= \texttt{<Var>()} \\
\texttt{<Letter>} &::= [a - zA - Z] \\
\texttt{<Var>} &::= \texttt{<Letter>} \mid \texttt{<Var>} \mathbin{+\!\!+} \texttt{<Letter>} \mid \texttt{<Var>} \mathbin{+\!\!+} \texttt{<Digit>}
\end{aligned}
$$